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Learning Whole-Body Humanoid Locomotion via Motion Generation and Motion Tracking

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***Abstract*—Whole-body humanoid locomotion is challenging due to high-dimensional control, morphological instability, and the need for real-time adaptation to various terrains using onboard perception. Directly applying reinforcement learning (RL) with reward shaping to humanoid locomotion often leads to lower-body-dominated behaviors, whereas imitation-based RL can learn more coordinated whole-body skills but is typically limited to replaying reference motions without a mechanism to adapt them online from perception for terrain-aware locomotion. To address this gap, we propose a whole-body humanoid locomotion framework that combines skills learned from reference motions with terrain-aware adaptation. We first train a diffusion model on retargeted human motions for real-time prediction of terrain-aware reference motions. Concurrently, we train a whole-body reference tracker with RL using this motion data. To improve robustness under imperfectly generated references, we further fine-tune the tracker with a frozen motion generator in a closed-loop setting. The resulting system supports directional goal-reaching control with terrain-aware whole-body adaptation, and can be deployed on a Unitree G1 humanoid robot with onboard perception and computation. The hardware experiments demonstrate successful traversal over boxes, hurdles, stairs, and mixed terrain combinations. Quantitative results further show the benefits of incorporating online motion generation and fine-tuning the motion tracker for improved generalization and robustness.**

I. Introduction

Developing whole-body perceptive humanoid locomotion remains a challenging yet important step toward expanding the operational range of humanoid robots. While deep reinforcement learning (RL) has achieved remarkable success in enabling highly dynamic behaviors on quadrupedal systems [1–6], extending these capabilities to humanoid robots remains considerably difficult, as humanoids possess higher degrees of freedom together with more complex morphological constraints. As a result, training an RL-based whole-body humanoid controller solely through reward shaping can be non-trivial and often suffers from inefficient exploration. Without explicit structural guidance, vanilla RL-based humanoid controllers frequently converge to locomotion strategies with limited whole-body coordination, making it difficult to acquire coordinated behaviors for challenging terrain traversal. This limitation becomes especially evident in tasks such as humanoid parkour, where coordinated whole-body movements are essential for traversing large and complex obstacles.

To alleviate the difficulty of reward shaping and inefficient exploration, motion imitation via RL has emerged as a promising alternative [7–12]. By leveraging reference data, motion tracking can efficiently transfer highly coordinated whole-body skills to the robot. However, pure motion trackers are inherently limited to replaying choreographed trajectories and generally lack the adaptability and reactivity required for operation in unstructured and diverse environments. Achieving human-level perceptive locomotion therefore requires a higher-level composition mechanism that can dynamically adjust locomotion style across varied obstacles based on real-time perceptive inputs.

One direct solution for skill composition is to distill multiple specialized teacher policies [13], or to compose multiple skills from separate reference trajectories within a single imitation-based teacher framework [14, 15]. By learning distinct expert skills for different terrain conditions, a unified distilled policy can learn to transition among them using exteroceptive observations. However, current methods have not yet demonstrated strong scalability to a diverse set of skills or robust generalization to new terrains.

Alternatively, generative models have recently shown strong potential for scalable motion control. Training generative models on large-scale motion datasets has proven effective in both character control [16–18] and robot control [9, 19], naturally reducing the engineering effort required as the motion repertoire expands. However, directly deploying expressive generated kinematic trajectories on real humanoid locomotion systems may introduce motion artifacts, such as foot sliding or temporal discontinuities, which can lead to catastrophic failure. To mitigate such issues, recent work [18, 20, 21] combines diffusion-based motion generation with RL-based motion tracking, allowing the tracker to output physically plausible actions based on kinematic generation. Despite this progress, their applicability to real humanoid robots in whole-body locomotion tasks remains insufficiently explored, and their deployment may potentially be hindered by slow inference and limited robustness to real-world disturbances.

To address these limitations, we propose a whole-body humanoid locomotion system that adapts a motion generation and motion tracking framework to real-time robot control for challenging humanoid locomotion tasks. In particular, we repurpose coordinated human whole-body skills learned from motion data for perceptive humanoid locomotion, and build a control framework that combines diffusion-based motion generation with RL-based motion tracking. The resulting system supports

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通过运动生成与运动跟踪学习全身人形机器人运动

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摘要—全身人形机器人运动具有挑战性，因为其控制维度高、形态不稳定，并且需要利用机载感知实时适应各种地形。将带奖励塑形的强化学习（RL）直接应用于人形机器人运动，往往会导致以下肢为主的行为；而基于模仿的 RL 虽能学习更协调的全身技能，但通常局限于回放参考运动，缺乏一种能够根据感知在线调整参考运动以实现地形感知运动的机制。为弥补这一空白，我们提出了一种全身人形机器人运动框架，将源自参考运动的技能与地形感知自适应相结合。我们首先在重定向后的人体运动上训练扩散模型，以实时预测地形感知的参考运动。同时，我们利用这些运动数据，通过 RL 训练一个全身参考跟踪器。为了提高在生成参考不完美情况下的鲁棒性，我们进一步在闭环设置中借助冻结的运动生成器微调该跟踪器。最终系统支持具有地形感知全身自适应的方向性目标到达控制，并可部署在配备机载感知与计算能力的宇树 G1 人形机器人上。硬件实验表明，该系统能够成功穿越箱子、跨栏、楼梯以及混合地形组合。定量结果进一步表明，引入在线运动生成并微调运动跟踪器有助于提升泛化性和鲁棒性。

I. 引言

开发全身感知的人形机器人运动能力，仍是扩展人形机器人作业范围过程中一个充满挑战但至关重要的步骤。尽管深度强化学习（RL）在使四足系统实现高动态行为方面已取得显著成功 [1–6], , 但将这些能力扩展到人形机器人仍然相当困难，因为人形机器人具有更高的自由度和更复杂的形态约束。因此，仅通过奖励塑形来训练基于RL的全身人形控制器可能并非易事，而且常常面临探索效率低下的问题。在没有显式结构指导的情况下，基础RL人形控制器往往会收敛到全身协调有限的行进策略，从而难以获得用于挑战性地形穿越的协调行为。这一局限在人形机器人跑酷等任务中尤为明显，在这些任务中，协调的全身运动对于穿越大型且复杂的障碍物至关重要。

为缓解奖励塑形的困难以及探索效率低下的问题，基于强化学习的运动模仿已成为一种有前景的替代方案 [7–12]。通过利用参考数据，运动跟踪能够高效地将高度协调的全身技能迁移到机器人上。然而，纯运动跟踪器本质上仅限于回放预先编排的轨迹，并且通常缺乏在非结构化和多样化环境中运行所需的适应性与反应能力。因此，实现人类水平的感知运动需要一种更高层次的组合机制，能够基于实时感知输入，在不同障碍物之间动态调整运动风格。

技能组合的一种直接解决方案是蒸馏多个专门的教师策略 [13], , 或在单个基于模仿的教师框架内，从独立的参考轨迹中组合多种技能 [14, 15]。通过为不同地形条件学习不同的专家技能，一个统一的蒸馏策略可以学会利用外感知观测在它们之间进行切换。然而，当前方法尚未展现出对多样化技能集的强大可扩展性，或对新地形的稳健泛化能力。

另一种思路是，生成模型近来在可扩展的运动控制方面展现出巨大潜力。在大规模运动数据集上训练生成模型已被证明有效在角色控制 [16–18] 和机器人控制 [9, 19], 中，随着动作库的扩展，自然地减少了所需的工程投入。然而，将富有表现力的生成运动学轨迹直接部署到真实的人形机器人运动系统上，可能会引入运动伪影，例如脚部滑动或时间不连续，这可能导致灾难性失败。为缓解此类问题，近期工作 [18, 20, 21] 将基于扩散的运动生成与基于 RL 的运动跟踪相结合，使跟踪器能够根据运动学生成输出物理上合理的动作。尽管取得了这些进展，它们在真实人形机器人全身运动任务中的适用性仍未得到充分探索，并且其部署可能会因推理速度慢以及对现实世界干扰的鲁棒性有限而受到阻碍。

为了解决这些局限，我们提出了一种全身人形机器人运动系统，该系统将运动生成与运动跟踪框架适配到实时机器人控制中，以应对具有挑战性的人形机器人运动任务。具体而言，我们重新利用从运动数据中学习到的协调人体全身技能，用于具备感知能力的人形机器人运动，并构建了一个将基于扩散的运动生成与基于强化学习的运动跟踪相结合的控制框架。由此得到的系统支持

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Fig. 1. Whole-body humanoid locomotion on the Unitree G1 robot enabled by diffusion-based motion generation and RL-based motion tracking. Our hardware experiments show behaviors including box climbing, box descent, vaulting, and traversal over mixed terrain sequences. https://wholebodylocomotion.github.io/.

directional goal-reaching control together with terrain-aware whole-body adaptation. Our hardware experiments show that our approach enables efficient skill execution and generalization across diverse terrains, including terrains unseen during training. Our results further demonstrate the effectiveness of using a motion generator trained on human motion data to guide an RL-based controller through challenging terrains requiring diverse skills, without heavily engineered distillation strategies.

II. Related work

A. Whole-body Motion Tracking for Humanoids

In simulated whole-body character control, prior works [7, 22] have explored RL-based frameworks in which characters learn to imitate reference motion clips. By training a control policy to minimize the discrepancy between the reference motion and the executed motion, the controller is able to reproduce a wide range of dynamic maneuvers while remaining consistent with the physical constraints of the controlled system.

More recently, this paradigm has been widely adopted in quadruped and humanoid robot control [9, 12, 23, 24]. By designing imitation rewards and regularization terms, and optimizing the policy with the PPO algorithm [25], recent works have demonstrated strong capability in imitating large-scale motion datasets and even enabling real-time teleoperation [11, 26, 27]. However, for general locomotion tasks, pure motion tracking remains fundamentally limited in its adaptability to varying terrains.

B. Motion Generation for Motion Control

Motion generation with generative models provides an effective way to sample feasible motion trajectories from curated motion datasets. Prior work [17] has demonstrated that diffusion models can generate smooth kinematic motion trajectories conditioned on text descriptions by incorporating additional geometric losses. Building on this line of work, Tevet et al. [18] further extend it to physical character control by coupling the generated motion with an RL-based motion tracker, which helps correct artifacts introduced by the generative model. Serifi et al. [20] also combine motion generation with RL-based motion tracking, and further fine-tune the generator using value information from the critic network of the tracking policy. Xu et al. [21] adopt a similar design philosophy to [18] and extend it to humanoid parkour in simulation, demonstrating impressive whole-body motion control conditioned on surrounding terrain observations. However, these methods mainly target simulated character control, and may still be constrained by the inference latency of diffusion models and lack of robustness for real robotic systems. As a result, relatively few real-time whole-body humanoid locomotion systems with integrated exteroceptive sensing have been demonstrated.

Another line of work [9, 16] generates joint state-action pairs without relying on a separate motion tracker. These methods have been demonstrated on humanoid robots for tasks such as waypoint navigation and motion inpainting. However, because control is produced directly from the co-diffusion process, they may still suffer from motion artifacts that can lead to unstable locomotion. Moreover, more dynamic and demanding tasks, such as humanoid whole-body locomotion, are not well-studied in this setting. In our work, we follow the general framework of [21], and adapt it to real-world humanoid locomotion tasks that require both whole-body coordination and real-time reaction to terrain conditions.

C. Skill Composition for Perceptive Locomotion Control

Composing multiple skills into a unified policy is often regarded as an effective shortcut toward whole-body visuomotor locomotion control, compared with training a single policy entirely from scratch. Hoeller et al. [28] first train multiple expert skills for quadrupedal locomotion across different terrain types, and then learn a high-level planner that switches categorically among the expert policies. In contrast, Rudin et al. [4] employ DAgger [29] to first distill several expert policies into a single policy, and subsequently fine-tune the distilled policy with RL on new terrains.

For humanoid whole-body locomotion, concurrent work [13] adopts a similar strategy to [4]: it first trains terrain-specific expert policies from scratch with RL and then distills them into a unified student policy. Wu et al. [14] likewise apply a similar distillation framework, but construct the expert skills from imitation-based RL policies trained with whole-body motion references. These methods have shown promising results for terrain traversal requiring coordinated whole-body control. However, both approaches may rely on carefully designed distillation procedures, such as expert assignment and data distribution design, to efficiently learn transitions and combinations across different skills.

In our work, instead of explicitly engineering a distillation pipeline over multiple expert skills, we use a diffusion-based

图1. 由基于扩散的运动生成和基于强化学习的运动跟踪所实现的 Unitree G1 机器人全身人形运动。我们的硬件实验展示了包括攀爬箱子、下箱、跨越以及穿越混合地形序列在内的行为。https://wholebodylocomotion.github.io/。

方向性目标到达控制，以及地形感知的全身适应。我们的硬件实验表明，我们的方法能够实现高效的技能执行，并泛化到多种地形，包括训练期间未见过的地形。我们的结果进一步证明了，使用在人体运动数据上训练的运动生成器来引导基于强化学习的控制器穿越需要多种技能的复杂地形是有效的，且无需高度工程化的蒸馏策略。

II. 相关工作

A. 人形机器人的全身运动跟踪

在仿真全身角色控制中，先前工作 [7,22] 探索了基于强化学习的框架，其中角色学习模仿参考运动片段。通过训练控制策略以最小化参考运动与执行运动之间的差异，控制器能够再现广泛的动态动作，同时保持与被控系统的物理约束一致。

最近，这一范式已被广泛应用于四足机器人和人形机器人控制 [9, 12, 23, 24]。通过设计模仿奖励和正则化项，并使用 PPO 算法优化策略， [25], 近期工作已展现出在模仿大规模运动数据集方面的强大能力，甚至能够实现实时遥操作 [11, 26, 27]。然而，对于通用运动任务，纯运动跟踪在适应变化地形方面仍存在根本性局限。

B. 面向运动控制的运动生成

基于生成模型的运动生成提供了一种有效方式，可采样可行的运动轨迹，来源为精心整理的运动数据集。先前工作 [17]已表明，扩散模型通过引入额外的几何损失，能够生成以文本描述为条件的平滑运动学运动轨迹。沿着这一研究方向，Tevet 等人 [18] 通过将生成的运动与基于 RL 的运动跟踪器耦合，进一步将其扩展到物理角色控制，这有助于纠正生成模型引入的伪影。Serifi 等人 [20] 还将运动生成与基于 RL 的运动跟踪相结合，并利用跟踪策略的评论家网络的价值信息进一步微调生成器。Xu 等人 [21]采用了与 [18]类似的设计理念，并将其扩展到仿真中的人形跑酷，展示了以周围地形观测为条件的令人印象深刻的全身运动控制。然而，这些方法主要面向仿真角色控制，并且可能仍受限于扩散模型的推理延迟，以及缺乏面向真实机器人系统的鲁棒性。因此，目前已展示的集成外部感知的实时全身人形运动系统相对较少。

另一类工作 [9, 16] 生成联合状态-动作对，而不依赖单独的运动跟踪器。这些方法已在人形机器人上针对路点导航和运动修复等任务得到验证。然而，由于控制直接由协同扩散过程产生，它们仍可能受到运动伪影的影响，进而导致不稳定的运动。此外，更具动态性和更高要求的任务（例如人形机器人全身运动）在这种设定下尚未得到充分研究。在我们的工作中，我们遵循 [21], 的总体框架，并将其适配到真实世界的人形机器人运动任务，这些任务既需要全身协调，也需要对地形条件做出实时反应。

C. 面向感知运动控制的技能组合

将多种技能组合成一个统一的策略，通常被视为实现全身视觉运动控制的一条有效捷径，相比完全从头训练单个策略。Hoeller 等人 [28] 首先针对不同地形类型训练多种四足运动专家技能，然后学习一个高层规划器，在专家策略之间进行分类式切换。相比之下，Rudin 等人 [4] 采用 DAgger [29] 先将多个专家策略蒸馏为一个策略，随后在新地形上使用强化学习（RL）对蒸馏后的策略进行微调。

对于人形全身运动，同期工作 [13] 采用了与 [4] 相似的策略：它首先使用 RL 从头训练针对特定地形的专家策略，然后将其蒸馏为一个统一的学生策略。Wu 等人 [14] 同样应用了类似的蒸馏框架，但其专家技能是基于使用全身运动参考训练得到的模仿式 RL 策略构建的。这些方法在需要协调全身控制的地形穿越任务中已展现出良好前景。然而，这两种方法可能都依赖于精心设计的蒸馏流程，例如专家分配和数据分布设计，以高效学习不同技能之间的过渡与组合。

在我们的工作中，我们并未显式设计一个跨多个专家技能的蒸馏流水线，而是使用基于扩散的

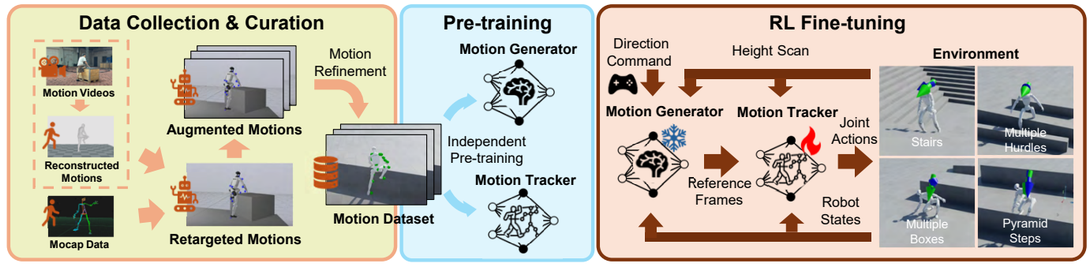

Fig. 2. Overview of the training framework. (a) Data Collection & Curation: whole-body robot motions are obtained from human motions through motion reconstruction and retargeting, followed by motion augmentation. (b) Pre-training: the diffusion-based motion generator and RL-based motion tracker are trained solely on the **offline motion dataset** from stage (a). (c) RL Fine-tuning: the motion tracker is **fine-tuned with RL** under randomized direction commands and on more diverse terrains with randomized configurations, including stairs, multiple vaulting hurdles, boxes, and pyramid-shaped high steps, while running the frozen motion generator.

motion generator trained on retargeted human motion data as a skill composition module that produces corresponding motion references for a low-level tracking policy based on the environment conditions. In addition, to reduce the impact of motion artifacts introduced by the generator and improve control performance, we further fine-tune the motion tracking policy with RL under randomized direction commands and more diverse terrain configurations while using a frozen motion generator pretrained on the offline motion dataset. More details are provided in Sec. III-C.

III. Method

The overall framework is illustrated in Fig. 2. Our method consists of three stages: (a) collecting whole-body motion data for different terrains and converting them into dynamically feasible robot trajectories; (b) pre-training the RL-based motion tracker and the diffusion-based motion generator on this dataset; and (c) fine-tuning the motion tracker on more diverse terrain configurations, while keeping the motion generator frozen and running it in a receding-horizon manner during deployment. We describe each stage below.

A. Data Collection & Curation

Our initial motion data consist of approximately 5 minutes of motion clips collected from two sources: motion videos captured by ourselves and public motion datasets [12, 30]. The dataset contains one representative motion type for each terrain skill, namely climbing onto a 50 cm box, vaulting over a 35 cm hurdle, jumping down from a 50 cm box, and stair ascent and descent on steps of approximately 20 cm height, together with omnidirectional walking motions such as straight walking and turning. To enlarge this limited initial dataset and improve terrain diversity, we further apply motion augmentation. In total, the resulting dataset contains approximately one hour of motion data for pre-training the motion generator and tracker.

1) Initial Dataset Construction: For motion skills recorded from videos, we first reconstruct human motions from raw videos using GVHMR [31]. We then retarget the reconstructed motions both from videos and motion datasets to the humanoid robot using a contact-constrained IK solver [32], similar to [12]. After obtaining the retargeted robot motions, we manually place terrain objects to align with the original motions and preserve the physical interaction between the robot and the environment. To ensure that the final training data are physically plausible for both the motion tracker and the motion generator, we do not directly use the raw retargeted trajectories; instead, we refine training motions from a DeepMimic-style tracking policy trained on those retargeted motions.

2) Motion Augmentation: To further improve the generalization of both the motion tracker and the motion generator, we enlarge the dataset through kinematics-based augmentation. Specifically, we augment existing motions by varying terrain geometry, including scaling obstacle heights and inserting random small boxes along the motion path, and then optimizing the motions to remain consistent with the modified terrain. The optimization includes losses such as terrain penetration and motion smoothness losses to reduce collisions and discontinuities [21]. As in the initial dataset construction, we do not use the optimized trajectories directly; instead, we train a tracking policy on them and record the resulting physically feasible trajectories. As a result, the augmented dataset covers box climbing and jump-down motions on 35–75 cm boxes, vaulting motions over 25–45 cm hurdles, stair motions on steps of 15–20 cm height, and omnidirectional walking motions with randomly placed small boxes.

B. Pre-training Stage

The pre-training stage equips both the motion generator and the motion tracker with the basic ability to react to different environments and to output appropriate motion references and joint actions.

1) Whole-body Motion Tracker: We first train a single whole-body motion tracker on a single RTX 4090 GPU using a DeepMimic-style RL framework with PPO in IsaacLab [33], based on the motion data collected in Sec. III-A. We design imitation rewards r_{mimic} and regularization terms r_{reg} (see Table I) so that the policy can imitate all reference motions while respecting the physical constraints of the real hardware:

$$R_{\text{pre}} = r_{\text{mimic}} + r_{\text{reg}}. \tag{1}$$

The tracker observation consists of the reference state, including linear and angular velocity in $\mathbb{R}^3$, joint position

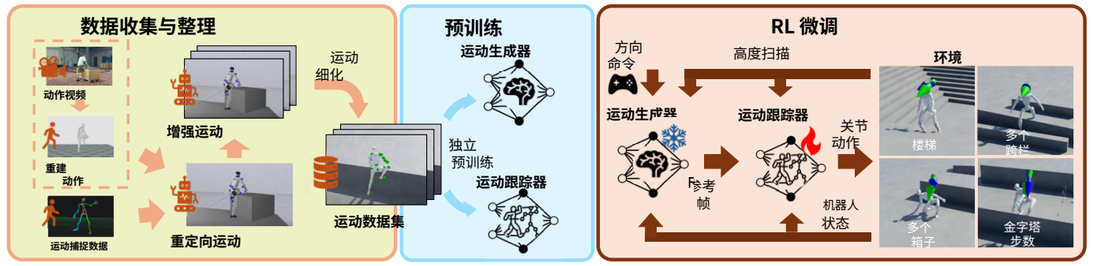

图 2. 训练框架概览。(a) 数据收集与整理：通过运动重建与重定向从人体运动中获取全身机器人运动，随后进行运动增强。(b) 预训练：基于扩散的运动生成器和基于强化学习的运动跟踪器仅在来自阶段 (a) 的 **离线运动数据集** 上进行训练。(c) 强化学习微调：在随机方向指令下，并在具有随机配置的更多样化地形上（包括楼梯、多个跨越障碍、箱子和金字塔形高台阶），同时运行冻结的运动生成器，对运动跟踪器 **使用强化学习进行微调**。

运动生成器，其基于重定向后的人体运动数据训练，作为一个技能组合模块，根据环境条件为低层跟踪策略生成相应的运动参考。此外，为降低生成器引入的运动伪影的影响并提升控制性能，我们在随机方向指令和更多样化的地形配置下，通过强化学习进一步微调运动跟踪策略，同时使用基于离线运动数据集预训练的冻结运动生成器。更多细节见第 III-C 节。

III. 方法

整体框架如图 2 所示。我们的方法包括三个阶段：(a) 收集不同地形的全身运动数据，并将其转换为动态可行的机器人轨迹；(b) 在该数据集上预训练基于强化学习的运动跟踪器和基于扩散的运动生成器；以及 (c) 在更多样化的地形配置上微调运动跟踪器，同时保持运动生成器冻结，并在部署期间以滚动时域方式运行它。下面我们分别描述每个阶段。

A. 数据收集与整理

我们的初始运动数据由约 5 分钟的运动片段组成，这些片段收集自两个来源：我们自己拍摄的运动视频以及公开运动数据集 [12, 30]。该数据集为每种地形技能包含一种代表性运动类型，即爬上 50 cm 箱子、跨越 35 cm 跨栏、从 50 cm 箱子跳下，以及在约 20 cm 高的台阶上上下楼梯，此外还包括直行和转向等全向行走运动。为了扩充这一有限的初始数据集并提高地形多样性，我们进一步应用了运动增强。最终，所得数据集包含约一小时的运动数据，用于预训练运动生成器和跟踪器。

1) 初始数据集构建：对于从视频中记录的运动技能，我们首先使用 GVHMR [31] 从原始视频中重建人体运动。随后，我们使用接触约束的 IK 求解器 [32], , 类似于 [12], 将来自视频和运动数据集的已重建运动重定向到人形机器人。获得重定向后的机器人运动后，我们手动放置地形对象，使其与原始运动对齐，并保持机器人与环境之间的物理交互。为确保最终训练数据对运动跟踪器和运动生成器都具有物理合理性，我们不直接使用原始重定向轨迹；相反，我们通过在那些重定向运动上训练的 DeepMimic 风格跟踪策略来精炼训练运动。

2) 动作增强：为了进一步提升运动跟踪器和运动生成器的泛化能力，我们通过基于运动学的增强来扩充数据集。具体而言，我们通过改变地形几何来增强现有动作，包括缩放障碍物高度以及沿运动路径插入随机小箱子，然后优化动作以与修改后的地形保持一致。优化包括地形穿透损失和运动平滑性损失等损失项，以减少碰撞和不连续现象 [21]。与初始数据集构建一样，我们并不直接使用优化后的轨迹；相反，我们在这些轨迹上训练跟踪策略，并记录由此得到的物理可行轨迹。因此，增强数据集涵盖 35–75 cm 箱子上的爬箱和跳下动作、25–45 cm 栏架上的跨越动作、15–20 cm 高台阶上的楼梯动作，以及带有随机放置小箱子的全向行走动作。

B. 预训练阶段

预训练阶段使运动生成器和运动跟踪器均具备对环境作出反应并输出合适运动参考与关节动作的基本能力。

1) 全身运动跟踪器：我们首先基于第 III-A 节收集的运动数据，在单个 RTX 4090 GPU 上，使用 IsaacLab [33], 中带 PPO 的 DeepMimic 风格强化学习框架训练一个单一全身运动跟踪器。我们设计了模仿奖励 r_{mimic} 和正则化项 r_{reg} （见表 I），以使策略在遵守真实硬件的物理约束的同时，能够模仿所有参考运动：

$$R_{\text{pre}} = r_{\text{mimic}} + r_{\text{reg}}. \tag{1}$$

跟踪器观测由参考状态组成，包括 $\mathbb{R}^3$ 中的线速度和角速度、关节位置

and velocity in $\mathbb{R}^{23}$, and key body positions in the base frame in $\mathbb{R}^{9\times3}$; proprioceptive terms, including five consecutive frames of base angular velocity, projected gravity in $\mathbb{R}^3$, joint position and velocity, and the previous action in $\mathbb{R}^{23}$; together with terrain height scans in $\mathbb{R}^{17\times11}$. The policy outputs 23-dimensional target joint positions as actions. Although Yang et al. [12] show that terrain information is not strictly required when the demonstrations are sufficiently accurate, we find that it can be beneficial to include exteroceptive input already at this stage, as it helps the later fine-tuning stage.

2) Diffusion-based Motion Generator: Our motion generator is based on a diffusion model that predicts whole-body reference motion sequences, following the architecture proposed in MDM [17, 21]. The model predicts future motion features over a 0.5-second horizon (25 frames), including root position in $\mathbb{R}^3$, root orientation in $\mathbb{R}^4$, joint positions in $\mathbb{R}^{23}$, and body link positions in $\mathbb{R}^{23\times3}$, conditioned on the target heading vector, terrain height scan, and the past two frames of motion features.

During training, we randomly sample motion sequences from the dataset, compute the heading vector from the base pose difference, and use the first two frames as conditions to predict the remaining frames. The reconstruction loss is then computed by comparing the predicted motion with the ground-truth sequence. In addition to the reconstruction loss, we incorporate geometric losses including velocity, joint consistency, and terrain penetration losses, similar to [21]. The generator is trained on the collected motion dataset and later serves as the motion planner that provides reference motions for the tracking policy over different terrains. To improve robustness against noisy robot states, we additionally perturb the height scans and previous-state conditions during training.

However, as discussed in Sec. II-B, generated motions may still contain artifacts, which can lead to failure when directly paired with a tracker trained mainly on high-quality references. Simply combining the pretrained motion generator and motion tracker can therefore result in limited robustness and reliability.

C. RL Fine-tuning Stage

In this stage, we improve the robustness and reliability of the pretrained motion generator and motion tracker, and adapt the full control system for real-time closed-loop deployment on a real humanoid robot. Directly combining the pretrained generator and tracker may still suffer from motion artifacts, tracking failures caused by imperfect references, and limited generalization to terrains and target directions beyond those covered by the training dataset. Therefore, we introduce an additional fine-tuning stage in which the motion tracker is fine-tuned with the frozen motion generator via RL on more diverse terrains and under varied target direction conditions.

During this stage, the motion generator produces reference frames conditioned on the past two frames of robot states to form a closed-loop motion prediction process, rather than relying on autoregressive feedback of its own previous predictions. To reduce deployment latency under limited onboard computation, we use only 2 denoising steps in the motion generator. We also inject additional noise into both the motion generator inference process and the tracker observations to improve robustness against real-world disturbances.

In addition, we randomize the target heading direction and introduce a heading tracking reward as a task reward r_{task} (see Table I), together with the imitation r_{mimic} and regularization terms r_{reg}, to encourage the full system to follow the desired direction while leveraging the motion priors acquired during pre-training:

$$R_{\text{post}} = r_{\text{mimic}} + r_{\text{reg}} + r_{\text{task}}. \tag{2}$$

We further introduce more diverse training terrains, including stairs with 15–25 cm step heights, continuous vaulting hurdles of 25–55 cm height, and multiple climbing boxes and pyramid steps ranging from 30–85 cm in height with varying widths as well as yaw and pitch angles.

After fine-tuning, the tracking policy effectively acts as a motion filter: it tracks the reference produced by the generator, while adjusting its execution to suppress unsafe behaviors using exteroceptive terrain observations. Despite the reduced number of denoising steps, which may further degrade raw generation quality, the fine-tuned tracker still robustly tracks the motion patterns produced by the generator and successfully traverses diverse terrains. Randomization of target headings and terrain configurations further enables behaviors beyond those present in the offline data, such as corner box climbing, reorientation on the box before descent, and continuous vaulting.

D. Details for Hardware Deployment

For hardware experiments, the entire pipeline is deployed fully onboard the Unitree G1 robot. We use DLIO [34] with the onboard Livox MID360 LiDAR and IMU to estimate the robot base pose, which is used as an input to motion generation. For terrain perception, we employ Elevation Mapping CuPy [35] together with the DLIO base pose to reconstruct terrain height information over different real-world environments.

Since the neck joint of the G1 robot is passive, aggressive motions can induce head movement that degrades base pose estimation. To improve state-estimation stability, we fuse the LiDAR IMU and torso IMU measurements by subtracting the orientation estimated by DLIO from the moving-average pitch measured by the torso IMU, which allows us to estimate the head pitch variation and compensate for its effect.

To accelerate diffusion-based motion generation, we deploy the generator with TensorRT, reducing inference time to approximately 0.02 s. Although the motion generator operates with a 0.5-second prediction horizon (2 Hz) during fine-tuning, we update the reference every 0.25 seconds in a receding-horizon way during deployment to improve responsiveness to the surrounding environment. The motion generator runs on a Jetson Thor mounted on the back of the robot, while the motion tracking controller and the remaining onboard modules run on the Jetson Orin integrated into the Unitree G1 platform.

IV. Experiments

In this section, we present the experimental results of the proposed humanoid locomotion system in simulation and on real hardware. We also provide quantitative evaluation to examine the contribution of motion generation to generalization, and the effect of RL fine-tuning of the motion tracker on robustness and success rates across terrain types.

以及 $\mathbb{R}^{23}$ 中的速度，以及 $\mathbb{R}^{9\times3}$ 中基座坐标系下的关键身体位置；本体感觉项，包括连续五帧的基座角速度、$\mathbb{R}^3$ 中的投影重力、关节位置和速度，以及 $\mathbb{R}^{23}$ 中的上一动作；还有 $\mathbb{R}^{17\times11}$ 中的地形高度扫描。策略输出 23 维目标关节位置作为动作。尽管 Yang 等人 [12] 表明，当示范足够准确时，地形信息并非严格必需，但我们发现，在此阶段就纳入外感受输入可能是有益的，因为它有助于后续的微调阶段。

2) 基于扩散的运动生成器：我们的运动生成器基于一个扩散模型，该模型遵循 MDM [17, 21] 中提出的架构，预测全身参考运动序列。该模型在 0.5 秒的时间范围（25 帧）内预测未来运动特征，包括 $\mathbb{R}^3$ 中的根位置、$\mathbb{R}^4$ 中的根朝向、$\mathbb{R}^{23}$ 中的关节位置以及 $\mathbb{R}^{23\times3}$ 中的身体连杆位置，并以目标朝向向量、地形高度扫描和过去两帧运动特征为条件。

在训练期间，我们从数据集中随机采样运动序列，根据基础姿态差异计算朝向向量，并使用前两帧作为条件来预测剩余帧。然后，通过将预测运动与真值序列进行比较来计算重建损失。除重建损失外，我们还引入了几何损失，包括速度、关节一致性和地形穿透损失，与 [21] 类似。生成器在收集的运动数据集上进行训练，随后作为运动规划器，为不同地形上的跟踪策略提供参考运动。为了提高对含噪机器人状态的鲁棒性，我们还在训练期间扰动高度扫描和先前状态条件。

然而，如第 II-B 节所述，生成的运动仍可能包含伪影，当其直接与主要基于高质量参考训练的跟踪器配合使用时，可能导致失败。因此，简单地将预训练的运动生成器和运动跟踪器组合起来，可能只能获得有限的鲁棒性和可靠性。

C. RL 微调阶段

在该阶段，我们提高预训练运动生成器和运动跟踪器的鲁棒性与可靠性，并使完整控制系统适应在真实人形机器人上的实时闭环部署。直接组合预训练的生成器和跟踪器仍可能面临运动伪影、由不完美参考导致的跟踪失败，以及对训练数据集未覆盖地形和目标方向的泛化能力有限等问题。因此，我们引入一个额外的微调阶段，在该阶段中，运动跟踪器在冻结运动生成器的情况下，通过 RL 在更多样化的地形以及不同目标方向条件下进行微调。

在此阶段，运动生成器基于机器人状态过去两帧生成参考帧，以形成闭环运动预测过程，而不是依赖其自身先前预测的自回归反馈。为了在有限的机载计算资源下降低部署延迟，我们在运动生成器中仅使用 2 步去噪。我们还向运动生成器推理过程和跟踪器观测中注入额外噪声，以提高对现实世界扰动的鲁棒性。

此外，我们随机化目标朝向，并引入朝向跟踪奖励作为任务奖励 r_{task}（见表 I），连同模仿 r_{mimic} 和正则化项 r_{reg}，以鼓励整个系统跟随期望方向，同时利用预训练期间获得的运动先验：

$$R_{\text{post}} = r_{\text{mimic}} + r_{\text{reg}} + r_{\text{task}}. \tag{2}$$

我们进一步引入更多样化的训练地形，包括台阶高度为 15–25 cm 的楼梯、高度为 25–55 cm 的连续跨栏障碍，以及高度在 30–85 cm 之间、宽度以及偏航角和俯仰角各不相同的多个攀爬箱和金字塔台阶。

微调后，跟踪策略实际上充当了运动滤波器：它跟踪生成器产生的参考，同时利用外部感知地形观测调整其执行，以抑制不安全行为。尽管去噪步数减少，这可能会进一步降低原始生成质量，但微调后的跟踪器仍能稳健地跟踪生成器产生的运动模式，并成功穿越各种地形。目标朝向和地形配置的随机化还进一步促成了离线数据中不存在的行为，例如攀爬角落箱子、下箱前在箱子上重新定向，以及连续翻越。

D. 硬件部署细节

对于硬件实验，整条流水线完全部署在 Unitree G1 机器人本体上。我们使用 DLIO [34]，结合机载 Livox MID360 激光雷达和 IMU 来估计机器人基座位姿，并将其作为运动生成的输入。对于地形感知，我们使用 Elevation Mapping CuPy [35]，并结合 DLIO 估计的基座位姿，在不同真实环境中重建地形高度信息。

由于G1机器人的颈部关节是被动关节，剧烈的运动可能会导致头部移动，从而降低基座位姿估计的准确性。为提高状态估计的稳定性，我们将LiDAR IMU和躯干IMU的测量数据进行融合：从躯干IMU测得的移动平均俯仰角中减去DLIO估计的姿态角，从而可以估计头部俯仰角的变化并补偿其影响。

为加速基于扩散的运动生成，我们使用TensorRT部署生成器，将推理时间缩短至约0.02秒。尽管在微调期间运动生成器以0.5秒的预测时域（2 Hz）运行，但在部署期间，我们以滚动时域的方式每0.25秒更新一次参考，以提高对周围环境的响应能力。运动生成器运行在安装在机器人背部的Jetson Thor上，而运动跟踪控制器和其余机载模块则运行在集成于Unitree G1平台的Jetson Orin上。

IV. 实验

在本节中，我们展示了所提出的人形运动系统在仿真和真实硬件上的实验结果。我们还提供了定量评估，以检验运动生成对泛化能力的贡献，以及运动跟踪器的强化学习微调对不同地形类型下鲁棒性和成功率的影响。

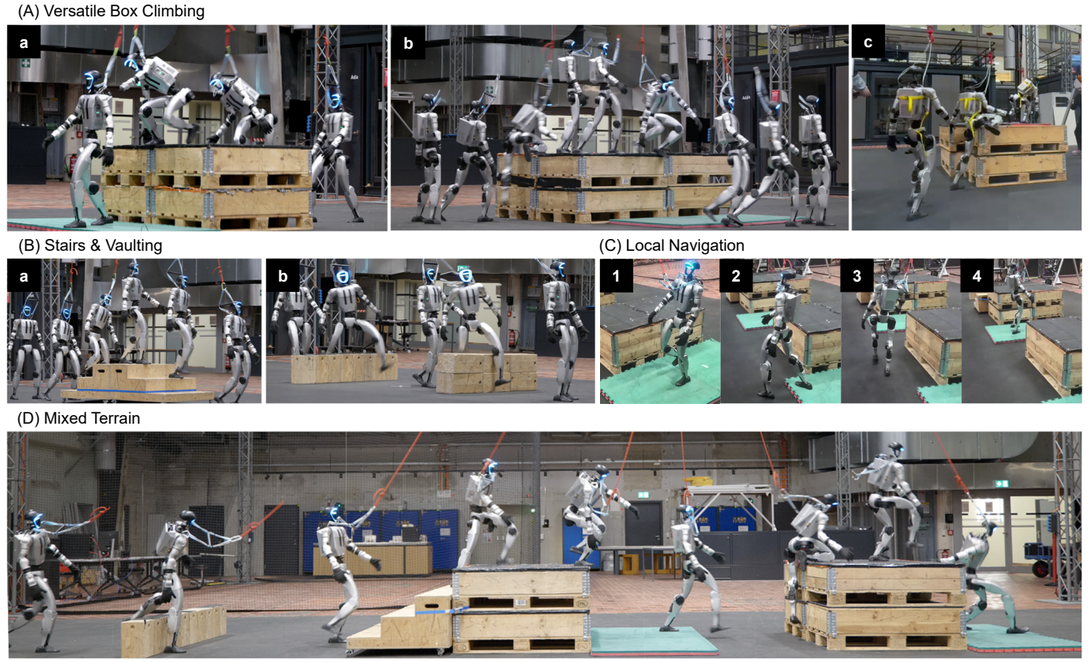

Fig. 3. Results of hardware experiments. (A) The robot climbs onto a 75 cm box and jumps down in three different ways: (a) straight ascent and descent; (b) straight ascent followed by a 90° right turn and side jump-down; and (c) ascent and descent from the box corner. (B) The robot traverses (a) a staircase and (b) a sequence of vaulting hurdles. (C) The robot shows local navigation behavior by bypassing the box to reach the target position. (D) The robot traverses a mixed terrain sequence consisting of a vaulting obstacle, stairs, and box climbing.

TABLE I
REWARD EQUATIONS FOR RL-BASED MOTION TRACKER

Group	Name	Equation
r_{mimic}	Base Pose Tracking	$3\exp\left(-4\left\|\mathbf{p}_b^w - \mathbf{p}_b^{w*}\right\|^2 - \left\|\mathbf{q}_b \otimes \mathbf{q}_b^{*-1}\right\|^2\right)$
	Base Velocity Tracking	$\exp\left(-\left\|\mathbf{v}_b^b - \mathbf{v}_b^{b*}\right\|^2 - 0.1\left\|\boldsymbol{\omega}_b^b - \boldsymbol{\omega}_b^{b*}\right\|^2\right)$
	Joint Position Tracking	$2.5\exp(-2\sum_{i=1}^{23} 0.25(q_i - q_i^*)^2)$
	Joint Velocity Tracking	$0.5\exp(-2\sum_{i=1}^{23} 0.01(\dot{q}_i - \dot{q}_i^*)^2)$
	Body Pos Tracking (Base)	$2\exp\left(-\sum_{i=1}^{9}\left\|\mathbf{p}_{l,i}^b - \mathbf{p}_{l,i}^{b*}\right\|^2\right)$
	Body Pos Tracking (Global)	$\exp\left(-\sum_{i=1}^{9}\left\|\mathbf{p}_{l,i}^w - \mathbf{p}_{l,i}^{w*}\right\|^2\right)$
r_{reg}	Action Rate (First-Order)	$-0.1\sum_{i=1}^{23}(a_{t,i} - a_{t-1,i})^2$
	Action Rate (Second-Order)	$-0.05\sum_{i=1}^{23}(a_{t,i} - 2a_{t-1,i} + a_{t-2,i})^2$
	Joint Position Limits	$-10\sum_{i=1}^{23}\max(q_i - q_{\lim,i}, 0)$
	Joint Velocity Limits	$-\sum_{i=1}^{23}\max(\dot{q}_i - \dot{q}_{\lim,i}, 0)$
	Torque Limits	$-0.001\sum_{i=1}^{23}\max(\tau_i - \tau_{\lim,i}, 0)$
	Joint Torques	$-0.00001\sum_{i=1}^{23}\tau_i^2$
	Body Linear Acceleration	$-0.01\sum_{i=1}^{9}\left\|\ddot{\mathbf{p}}_{l,i}^w\right\|^2$
r_{task}	Target Heading Tracking	$1.5\langle\mathbf{v}_b^b, \mathbf{d}_b\rangle / \left\|\mathbf{v}_b^b\right\|$

TABLE II
SYMBOLS USED IN TABLE I

Symbol	Description
$(\cdot)^*$	Reference quantity
$(\cdot)^b$	Expressed in the base frame
$(\cdot)^w$	Expressed in the world frame
$\mathbf{p}_b$	Base position
$\mathbf{q}_b$	Base orientation quaternion
$\mathbf{v}_b$	Base linear velocity
$\boldsymbol{\omega}_b$	Base angular velocity
$\mathbf{d}_b$	Base target heading
$\mathbf{p}_{l,i}$	Position of key body i
$\ddot{\mathbf{p}}_{l,i}$	Acceleration of key body i
q_i	Position of joint i
$\dot{q}_i$	Velocity of joint i
τ_i	Torque of joint i
$q_{\lim,i}$	Joint position limit of joint i
$\dot{q}_{\lim,i}$	Joint velocity limit of joint i
$\tau_{\lim,i}$	Joint torque limit of joint i

A. Hardware Results

We evaluate the proposed pipeline on real hardware as shown in Fig. 1, 3, and the supplementary video. The hardware experiments demonstrate both the effectiveness of our controller and its generalization ability across diverse terrains. In particular, we evaluate the controller on a range of terrain types, including box ascent and descent, stair ascent and descent, and continuous vaulting over obstacles of varying heights, as well as on more challenging compound terrain settings that combine vaulting, stairs, and box climbing within a single run. For box ascent and descent, our controller exhibits versatile climbing behaviors, including box traversal from different approach directions as well as the ability to traverse a box with on-top reorientation before descent. During ascent, the robot uses its knees and hands for support, while during descent it jumps off the box and uses the hands for cushioning, consistent with the motion style observed in the training data. For vaulting, the robot can traverse multiple hurdles of different heights, typically crossing over them directly rather than stepping onto the hurdle first.

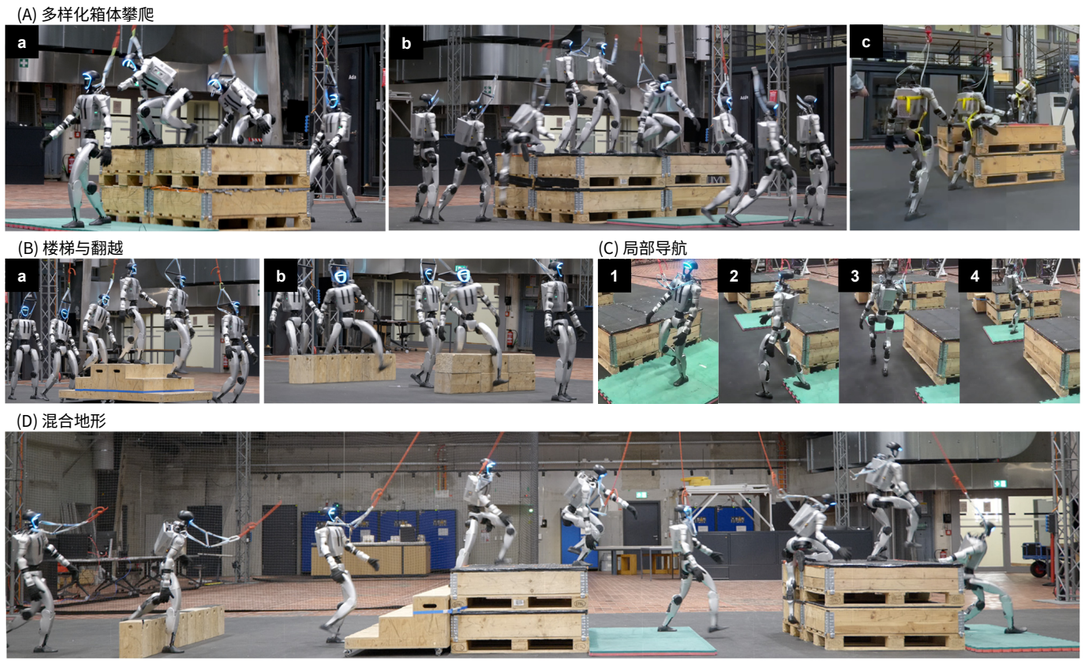

图 3. 硬件实验结果。（A）机器人爬上 75 厘米高的箱子，并以三种不同方式跳下：（a）直上直下；（b）直上后完成一次 90° 右转并侧向跳下；（c）从箱子角部上箱并跳下。（B）机器人穿越（a）一段楼梯和（b）一连串翻越栏架。（C）机器人通过绕过箱子到达目标位置，展示了局部导航行为。（D）机器人穿越由翻越障碍、楼梯和爬箱组成的混合地形序列。

表 I 基于强化学习的运动跟踪器奖励方程

组别	Name	方程
$r_{\text{模仿}}$	基座位姿跟踪	$3\exp\left(-4\left\|\mathbf{p}_b^w - \mathbf{p}_b^{w*}\right\|^2 - \left\|\mathbf{q}_b \otimes \mathbf{q}_b^{*-1}\right\|^2\right)$
	基座速度跟踪	$\exp\left(-\left\|\mathbf{v}_b^b - \mathbf{v}_b^{b*}\right\|^2 - 0.1\left\|\boldsymbol{\omega}_b^b - \boldsymbol{\omega}_b^{b*}\right\|^2\right)$
	关节位置跟踪	$2.5\exp(-2\sum_{i=1}^{23} 0.25(q_i - q_i^*)^2)$
	关节速度跟踪	$0.5\exp(-2\sum_{i=1}^{23} 0.01(\dot{q}_i - \dot{q}_i^*)^2)$
	身体位置跟踪（基座）	$2\exp\left(-\sum_{i=1}^{9}\left\|\mathbf{p}^b_{,i} - \mathbf{p}^{b*}_{,i}\right\|^2\right)$
	身体位置跟踪（全局）	$\exp\left(-\sum_{i=1}^{9}\left\|\mathbf{p}^w_{,i} - \mathbf{p}^{w*}_{,i}\right\|^2\right)$
r_{reg}	动作变化率（一阶）	$-0.1\sum_{i=1}^{23}(a_{t,i} - a_{t-1,i})^2$
	动作速率（二阶）	$-0.05\sum_{i=1}^{23}(a_{t,i} - 2a_{t-1,i} + a_{t-2,i})^2$
	关节位置限制	$-10\sum_{i=1}^{23}\max(q_i - q_{\lim,i}, 0)$
	关节速度限制	$-\sum_{i=1}^{23}\max(\dot{q}_i - \dot{q}_{\lim,i}, 0)$
	力矩限制	$-0.001\sum_{i=1}^{23}\max(\tau_i - \tau_{\lim,i}, 0)$
	关节力矩	$-0.00001\sum_{i=1}^{23}\tau_i^2$
	身体线性加速度	$-0.01\sum_{i=1}^{9}\left\|\ddot{\mathbf{p}}_{l,i}^w\right\|^2$
r_{task}	目标航向跟踪	$1.5\langle\mathbf{v}_b^b, \mathbf{d}_b\rangle / \left\|\mathbf{v}_b^b\right\|$

表 II 表 I 中使用的符号

符号	描述
$(\cdot)^*$	参考量
$(\cdot)^b$	在基坐标系中表示
$(\cdot)^w$	在世界坐标系中表示
$\mathbf{p}_b$	基座位置
$\mathbf{q}_b$	基座姿态四元数
$\mathbf{v}_b$	基座线速度 y
$\boldsymbol{\omega}_b$	基座角速度
$\mathbf{d}_b$	基座目标朝向
$\mathbf{p}_{l,i}$	关键身体部位 i 的位置
$\ddot{\mathbf{p}}_{l,i}$	关键身体 i 的加速度
q_i	关节 i的位置
$\dot{q}_i$	关节 i 的速度
τ_i	关节 i 的力矩
$q_{\lim,\ i}$	关节 i的关节位置限位
$\dot{q}_{\lim,\ i}$	关节 i 的关节速度限制
$\tau_{\lim,i}$	关节 i 的关节力矩限制

A. 硬件结果

我们在真实硬件上评估了所提出的流程，如图1、图3和补充视频所示。硬件实验既证明了我们控制器的有效性，也证明了其在多样地形上的泛化能力。特别地，我们在一系列地形类型上评估控制器，包括上下箱体、上下楼梯、连续跨越不同高度的障碍物，以及在更多地形上具有挑战性的复合地形设置，将跨越、楼梯和箱体攀爬结合在单次运行中。对于箱体上升和下降，我们的控制器表现出多样化的攀爬行为，包括从不同接近方向穿越箱体，以及在下降前在箱体顶部重新定向后穿越箱体的能力。在上升过程中，机器人使用膝盖和手进行支撑，而在下降过程中，它从箱体跳下并用手进行缓冲，这与训练数据中观察到的运动风格一致。对于跨越，机器人可以穿越多个不同高度的障碍物，通常直接越过它们，而不是先踩到障碍物上。

TABLE III
COMPARATIVE SUCCESS RATES OF FIXED-REFERENCE MOTION TRACKING AND THE PROPOSED FULL SYSTEM ACROSS THREE EVALUATION SETTINGS.

(a) Box Climbing

Box Height [cm]	Yaw [deg]	Tracker Only	Tracker+Gen.
40	0	0.982	1.000
50	-30	0.988	0.980
	30	1.000	0.982
	-15	0.974	1.000
	15	0.970	0.998
	0	0.998	1.000
60	0	0.984	0.996
70	0	0.606	0.966
80	0	0.230	0.962
Mean ± SD		0.859 ± 0.252	**0.987 ± 0.014**

(b) Vaulting

Hurdle Height [cm]	Yaw [deg]	Tracker Only	Tracker+Gen.
25	0	0.978	1.000
35	-40	0.808	1.000
	40	0.492	1.000
	-20	0.922	1.000
	20	0.954	1.000
	0	0.988	1.000
45	0	0.950	1.000
55	0	0.348	0.920
Mean ± SD		0.805 ± 0.231	**0.990 ± 0.026**

(c) Ascending Stairs

Step Height [cm]	Yaw [deg]	Tracker Only	Tracker+Gen.
15	0	1.000	1.000
20	-30	0.948	1.000
	30	0.916	0.994
	-15	0.970	1.000
	15	0.990	0.996
	0	0.980	1.000
25	0	0.114	0.986
Mean ± SD		0.845 ± 0.300	**0.997 ± 0.005**

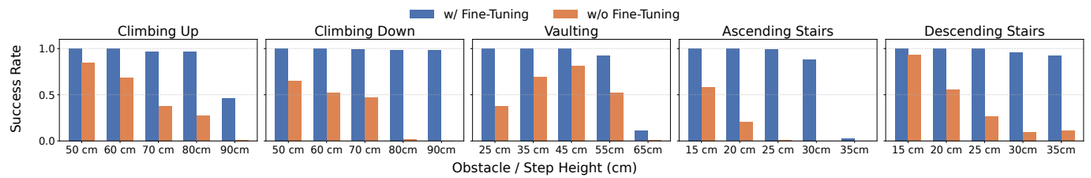

Fig. 4. Comparative results of the success rates of the motion tracker with and without fine-tuning across five terrain traversal tasks, including climbing up/down, vaulting, and stair ascent/descent. For each task, we spawn 500 robots in simulation with randomized initial poses and assign each of them a goal position that guides traversal of the corresponding terrain obstacle, from which the target direction is computed. The results consistently show that fine-tuning the tracker with a frozen motion generator improves the success rate of reaching the goal.

For stairs, the controller switches to stair ascent and descent behaviors. When these terrain types are combined, the controller can also transition dynamically among different motion styles according to the terrain, including sequences such as stair ascent following a jump-down motion, even though such terrain combinations are not seen during RL fine-tuning.

In addition, as shown in Fig. 3C and the supplementary video, we observe a form of local navigation behavior in the system after RL fine-tuning: when the robot is not in a suitable state to traverse an obstacle under the direction command computed from the given goal position using the motion generated by the planner, the tracking policy may partially override the reference motion and instead bypass the obstacle from the side to avoid failure and still reach the goal.

B. Quantitative Results

We further analyze two key design choices of our framework through quantitative simulation results: whether online motion generation improves adaptation and generalization across diverse terrains at deployment, and whether the additional fine-tuning stage is necessary after coupling the generator with the motion tracker. As shown in Table III and Fig. 4, both components are important for the final system.

1) Benefit of Online Motion Generation: We first evaluate whether online motion generation is beneficial for enabling the final system to adapt to diverse terrain variations by comparing it with fixed-reference tracking. We compare two settings on three representative tasks: box climbing, vaulting, and stair ascent. For box climbing, the reference trajectory is a motion for climbing onto a 50 cm box; for vaulting, the reference is a trajectory for traversing a 35 cm-high and 20 cm-wide hurdle; for stair ascent, the reference is a trajectory for ascending a staircase with 30 cm-wide and 20 cm-high steps. We then modify the terrain geometry at test time by changing the obstacle height or rotating the terrain mesh in yaw. **Tracker Only** uses the fine-tuned motion tracker with a fixed reference trajectory, and success is measured by whether the robot reaches the final XY base position of the reference trajectory. **Tracker + Gen.** uses the proposed full system, where the motion generator produces reference motions online from the target direction and terrain input, and success is measured by whether the robot reaches the goal position placed on the terrain. The goal position is chosen to match the final base position of the reference trajectory used in the *Tracker Only* setting, and the direction command is computed online accordingly. For each task, 500 robots are spawned in simulation with randomized initial poses and joint positions.

As shown in Table III, the full system with online motion generation achieves better generalization and higher average success rates across all three tasks than fixed-reference tracking on the corresponding terrain settings. For box climbing, the *Tracker Only* baseline performs well close to the nominal reference condition, but becomes brittle as the obstacle height increases. In contrast, *Tracker + Gen.* remains consistently robust across all tested settings. For vaulting, the fine-tuned tracker already exhibits some local generalization and can handle different hurdle heights, but its performance still degrades under larger changes in terrain height and orientation. The full system, by comparison, remains stable across all evaluated cases. A similar pattern appears in stair ascent: while the fixed-reference tracker can tolerate yaw perturbations, it deteriorates markedly under the height increase, whereas the full system continues to achieve nearly perfect performance.

These results highlight two points. First, the fine-tuned tracker is not entirely rigid. Because it observes terrain height scans and is fine-tuned on diverse terrain configurations, it can retain a limited degree of local terrain generalization. This is particularly visible in vaulting and box climbing, where nearby obstacle scales can sometimes be handled

表 III 固定参考运动跟踪与所提完整系统在三种评估设置下的成功率对比。

(a) 箱体攀爬

箱体高度 [cm]	Yaw [度]	仅跟踪器	Tracker+Gen.
40	0	0.982	1.000
50	-30	0.988	0.980
	30	1.000	0.982
	-15	0.974	1.000
	15	0.970	0.998
	0	0.998	1.000
60	0	0.984	0.996
70	0	0.606	0.966
80	0	0.230	0.962
均值 ± 标准差		0.859 ± 0.252	**0.987 ± 0.014**

(b) 翻越

跨栏高度 [cm]	Yaw [度]	仅跟踪	跟踪+生成
25	0	0.978	1.000
35	-40	0.808	1.000
	40	0.492	1.000
	-20	0.922	1.000
	20	0.954	1.000
	0	0.988	1.000
45	0	0.950	1.000
55	0	0.348	0.920
均值 ± 标准差		0.805 ± 0.231	**0.990 ± 0.026**

(c) 上楼梯

台阶高度 [cm]	Yaw [度]	仅跟踪器	跟踪器+生成
15	0	1.000	1.000
20	-30	0.948	1.000
	30	0.916	0.994
	-15	0.970	1.000
	15	0.990	0.996
	0	0.980	1.000
25	0	0.114	0.986
均值 ± 标准差		0.845 ± 0.300	**0.997 ± 0.005**

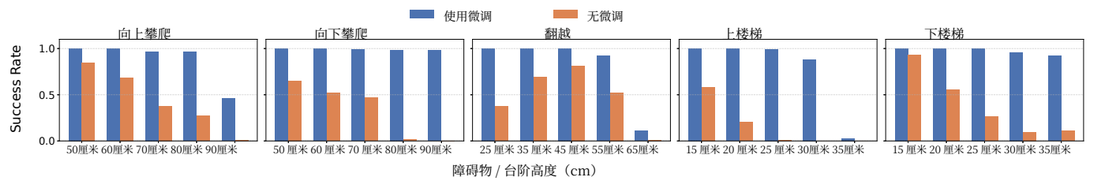

图4. 在五种地形穿越任务（包括向上/向下攀爬、翻越以及上/下楼梯）中，有無微调的运动跟踪器成功率对比结果。对于每个任务，我们在仿真中生成500个具有随机初始位姿的机器人，并为每个机器人分配一个目标位置，该位置引导其穿越相应的地形障碍，并据此计算目标方向。结果一致表明，使用冻结的运动生成器对跟踪器进行微调可提高到达目标的成功率。

对于楼梯，控制器切换到上楼和下楼行为。当这些地形类型组合在一起时，控制器还可以根据地形在不同运动风格之间动态转换，包括诸如跳跃下降后紧接着上楼这样的序列，即使在强化学习微调期间未见过此类地形组合。

此外，如图3C和补充视频所示，我们在RL微调后观察到系统中表现出一种局部导航行为：当机器人根据给定目标位置、利用规划器生成的运动计算出的方向指令下，并不处于适合穿越障碍物的状态时，跟踪策略可能会部分覆盖参考运动，转而从侧面绕开障碍物，以避免失败并仍能到达目标。

B. 定量结果

我们通过定量仿真结果进一步分析了本框架的两个关键设计选择：在部署时，在线运动生成是否能够提升在不同地形上的适应性和泛化能力；以及将生成器与运动跟踪器耦合后，额外的微调阶段是否必要。如表 III 和图 4 所示，这两个组件对最终系统都很重要。

1）在线运动生成的优势： 我们首先通过将其与固定参考轨迹跟踪进行比较，评估在线运动生成是否有助于使最终系统适应多样的地形变化。我们在三个代表性任务上比较两种设置：爬箱、翻越和爬楼梯。对于爬箱，参考轨迹是爬上一个 50 厘米高箱子的动作；对于翻越，参考轨迹是越过一个高 35 厘米、宽 20 厘米的栏架的动作；对于爬楼梯，参考轨迹是登上台阶宽 30 厘米、高 20 厘米的楼梯的动作。我们随后在测试时修改地形几何，方法是改变

障碍物高度，或者绕偏航方向旋转地形网格。**仅跟踪器** 使用经过微调的运动跟踪器和固定的参考轨迹，成功与否根据机器人是否到达参考轨迹的最终 XY 基座位置来衡量。**跟踪器 +生成** 使用所提出的完整系统，其中运动生成器根据目标方向和地形输入在线生成参考运动，成功与否根据机器人是否到达放置在地形上的目标位置来衡量。目标位置的选择与 仅跟踪器设置中所用参考轨迹的最终基座位置相匹配，方向指令也据此在线计算。对于每个任务，在仿真中生成 500 个机器人，其初始位姿和关节位置随机化。

如表 III 所示，在所有三项任务上，采用在线运动生成的完整系统均比相应地形设置下的固定参考跟踪具有更好的泛化能力和更高的平均成功率。对于爬箱任务，仅跟踪器基线在接近标称参考条件时表现良好，但随着障碍物高度增加会变得脆弱。相比之下，跟踪器 + 生成器在所有测试设置下始终保持稳健。对于翻越任务，微调后的跟踪器已表现出一定的局部泛化能力，能够处理不同的栏架高度，但其性能在地形高度和方向发生较大变化时仍会下降。相比之下，完整系统在所有评估情形下均保持稳定。爬楼梯任务中也出现了类似模式：固定参考跟踪器虽能容忍偏航扰动，但在高度增加时性能会显著恶化，而完整系统则继续实现近乎完美的性能。

这些结果凸显了两点。首先，微调后的跟踪器并非完全刚性。由于它观测地形高度扫描，并在多样化的地形配置上进行了微调，因此能够保留有限程度的局部地形泛化能力。这在撑越和爬箱时尤其明显，此时有时可以处理附近障碍物的尺寸

even without adapting the reference motion. Second, this limited generalization is fundamentally insufficient for robust perceptive locomotion. Once the terrain geometry changes more substantially, especially in ways that require changing the timing or style of the motion rather than merely tolerating small mismatch, the fixed-reference tracker degrades rapidly. During deployment, terrain-aware reference motions produced online by the motion generator allow the full system to better adapt its traversal behavior to diverse obstacle geometries. Overall, the results in Table III highlight the importance of online motion generation for adaptation and generalization across diverse terrains in the final system.

2) Impact of Fine-Tuning the Motion Tracker: The above results show that online motion generation is important for generalization during deployment. We next ask whether the additional fine-tuning stage is also necessary, or whether the pretrained tracker can simply be combined with the motion generator directly. To answer this question, we compare the pretrained tracker and the fine-tuned tracker under the same motion generator, as shown in Fig. 4. We evaluate five terrain traversal tasks, namely climbing up, climbing down, vaulting, ascending stairs, and descending stairs. For each task, a goal position is placed behind the terrain obstacle, from which the direction command is computed, and 500 robots are spawned in simulation to measure the success rate.

Figure 4 shows that fine-tuning the tracker consistently improves success rates across all evaluated tasks, with the largest gains appearing on the more difficult terrain settings. For climbing up and climbing down, the improvement becomes more apparent as the box height increases, indicating that fine-tuning is especially important when the robot is required to react robustly to more demanding vertical transitions. For stair ascent and descent, the fine-tuned tracker also shows clear advantages on larger step heights. Vaulting is comparatively easier than the other tasks, but the fine-tuned tracker still remains more reliable overall. Taken together, the histogram suggests that fine-tuning the tracker with RL does not merely provide marginal refinement, but systematically improves robustness across a broad range of traversal tasks.

A likely reason is that the pretrained tracker is optimized on a fixed set of offline motion trajectories, which are relatively smooth and physically filtered. During deployment, however, the online references generated by the motion generator are conditioned on noisy robot state, target direction, and terrain observations, and may therefore contain temporal discontinuities or motion artifacts. This creates a distribution mismatch between the tracker training references and the actual reference stream encountered during control. Fine-tuning with the frozen generator is therefore crucial: it allows the tracker to adapt to the distribution and imperfections of generated references, learn how to make effective use of motion patterns for task completion, and suppress unsafe behaviors that could otherwise lead to failure. This observation is also consistent with the qualitative behavior discussed in Sec. III-C, where the fine-tuned tracker effectively acts as a motion filter over the generated references. Overall, Fig. 4 shows that the fine-tuning stage is an important part of making the combined generation-and-tracking framework robust in practice.

V. Conclusion

Overall, our results demonstrate that combining motion generation with motion tracking is an effective way to realize perceptive whole-body humanoid locomotion. The proposed system enables efficient traversal over diverse terrain types and terrain combinations unseen during training, while the quantitative analysis highlights the importance of both online motion generation and tracker fine-tuning for adaptation and robustness.

One limitation of the current system is that it relies on a LiDAR-based elevation mapping pipeline to reconstruct the surrounding environment. When the mapping quality degrades due to sensing noises, the locomotion performance can deteriorate substantially. A promising direction for future work is to adopt perception architectures that do not depend as strongly on high-quality geometric sensing, such as neural mapping [6], or to incorporate a belief encoder [2] that compensates for sensing failures through proprioceptive observations. Another important direction is to extend the framework beyond pure locomotion to a broader range of loco-manipulation tasks and to more challenging outdoor environments.

Acknowledgments

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References

[1] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning quadrupedal locomotion over challenging terrain," *Science robotics*, vol. 5, no. 47, p. eabc5986, 2020.

[2] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning robust perceptive locomotion for quadrupedal robots in the wild," *Science Robotics*, vol. 7, no. 62, 2022.

[3] J. He, C. Zhang, F. Jenelten, R. Grandia, M. Bächer, and M. Hutter, "Attention-based map encoding for learning generalized legged locomotion," *Science Robotics*, vol. 10, no. 105, p. eadv3604, 2025.

[4] N. Rudin, J. He, J. Aurand, and M. Hutter, "Parkour in the wild: Learning a general and extensible agile locomotion policy using multi-expert distillation and rl fine-tuning," 2025.

[5] Z. Zhang, C. Li, T. Miki, and M. Hutter, "Motion priors reimagined: Adapting flat-terrain skills for complex quadruped mobility," 2025.

[6] C. Zhang, V. Klemm, F. Yang, and M. Hutter, "Ame-2: Agile and generalized legged locomotion via attention-based neural map encoding," 2026.

[7] X. B. Peng, P. Abbeel, S. Levine, and M. van de Panne, "Deepmimic: Example-guided deep reinforcement learning of physics-based character skills," *ACM Trans. Graph.*, vol. 37, no. 4, pp. 143:1–143:14, Jul. 2018.

即使不调整参考运动。其次，这种有限的泛化从根本上不足以实现稳健的感知运动。一旦地形几何发生更大幅度的变化，尤其是当这些变化要求改变运动的时机或风格，而不仅仅是容忍小失配时，固定参考跟踪器就会迅速退化。在部署期间，由运动生成器在线生成的地形感知参考运动使整个系统能够更好地调整其穿越行为，以适应多样化的障碍物几何形状。总体而言，表 III 中的结果凸显了在线运动生成对于最终系统在多样地形中实现适应与泛化的重要性。

2) 微调运动跟踪器的影响：上述结果表明，在线运动生成对于部署期间的泛化至关重要。接下来我们探讨额外的微调阶段是否也是必要的，或者预训练跟踪器是否可以直接与运动生成器结合。为回答这一问题，我们在相同运动生成器下比较预训练跟踪器和微调后的跟踪器，如图4所示。我们评估五项地形穿越任务，即向上攀爬、向下攀爬、翻越、上楼梯和下楼梯。对于每项任务，在地形障碍物后方放置一个目标位置，并据此计算方向指令，同时在仿真中生成500个机器人以测量成功率。

图4表明，对跟踪器进行微调在所有评估任务中都能持续提高成功率，其中在更困难的地形设置上提升最大。对于向上攀爬和向下攀爬，随着箱子高度增加，改进变得更加明显，这表明当机器人需要对更高难度的垂直过渡做出稳健反应时，微调尤为重要。对于上楼梯和下楼梯，微调后的跟踪器在更大的台阶高度上也显示出明显优势。翻越相较其他任务更容易，但微调后的跟踪器总体上仍然更可靠。综合来看，该直方图表明，使用RL对跟踪器进行微调并不仅仅带来边际改进，而是在广泛的地形穿越任务中系统性地提升了鲁棒性。

一个可能的原因是，预训练跟踪器是在一组固定的离线运动轨迹上优化的，这些轨迹相对平滑，并经过了物理过滤。然而在部署过程中，运动生成器生成的在线参考会受到带噪声的机器人状态、目标方向和地形观测的影响，因此可能包含时间上的不连续或运动伪影。这导致跟踪器训练时所使用的参考与在控制过程中实际遇到的参考流之间存在分布不匹配。因此，使用冻结的生成器进行微调至关重要：它使跟踪器能够适应生成参考的分布与不完美之处，学习如何有效利用运动模式来完成任务，并抑制否则可能导致失败的不安全行为。这一观察也与第 III-C 节中讨论的定性行为一致，其中微调后的跟踪器实际上充当了生成参考上的运动滤波器。总体而言，图 4 表明，微调阶段是使生成与跟踪联合框架在实践中保持稳健的重要组成部分。

V. 结论

总体而言，我们的结果表明，将运动生成与运动跟踪相结合，是实现具备感知能力的全身人形机器人运动的有效途径。所提出的系统能够高效穿越训练中未见过的多样地形类型和地形组合，而定量分析则凸显了在线运动生成和跟踪器微调对于适应性与稳健性的重要性。

当前系统的一个局限在于，它依赖基于 LiDAR 的高程建图流程来重建周围环境。当建图质量因传感噪声而下降时，运动性能可能会显著恶化。未来工作中一个有前景的方向是采用不那么强烈依赖高质量几何感知的感知架构，例如神经建图 [6]，或者引入一个通过本体感觉观测来补偿传感失效的信念编码器 [2]。另一个重要方向是将该框架从纯粹的移动扩展到更广泛的移动操作任务以及更具挑战性的户外环境。

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参考文献

[1] J. Lee、J. Hwangbo、L. Wellhausen、V. Koltun 与 M. Hutter，“在复杂地形上学习四足运动”，科学机器人，第5卷，第47期，文章编号 eabc5986，2020年。[2] T. Miki、J. Lee、J. Hwangbo、L. Wellhausen、V. Koltun 与 M. Hutter，“在野外为四足机器人学习鲁棒感知运动”，科学机器人，第7卷，第62期，2022年。[3] J. He、C. Zhang、F. Jenelten、R. Grandia、M. Bächer 与 M. Hutter，“基于注意力的地图编码用于学习泛化腿式运动”，科学机器人，第10卷，第105期，文章编号 eadv3604，2025年。[4] N. Rudin、J. He、J. Aurand 与 M. Hutter，“野外跑酷：利用多专家蒸馏和强化学习微调学习通用且可扩展的敏捷运动策略”，2025年。[5]Z. Zhang、C. Li、T. Miki 与 M. Hutter，“重新构想运动先验：将平地技能适配于复杂四足移动”，2025年。[6] C. Zhang、V. Klemm、F. Yang 与 M. Hutter，“Ame-2：通过基于注意力的神经地图编码实现敏捷且泛化的腿式运动”，2026年。[7] X. B. Peng、P. Abbeel、S. Levine 与 M. van de Panne，“Deepmimic：示例引导的基于物理的角色技能深度强化学习”，*ACM* 图形学汇刊，第37卷，第4期，第143:1–143:14页，2018年7月。

[8] Z. Chen, M. Ji, X. Cheng, X. Peng, X. B. Peng, and X. Wang, "Gmt: General motion tracking for humanoid whole-body control," *arXiv:2506.14770*, 2025.

[9] Q. Liao, T. E. Truong, X. Huang, Y. Gao, G. Tevet, K. Sreenath, and C. K. Liu, "Beyondmimic: From motion tracking to versatile humanoid control via guided diffusion," 2025.

[10] M. Ji, X. Peng, F. Liu, J. Li, G. Yang, X. Cheng, and X. Wang, "Exbody2: Advanced expressive humanoid whole-body control," *arXiv preprint arXiv:2412.13196*, 2024.

[11] Y. Ze, Z. Chen, J. P. Araújo, Z. ang Cao, X. B. Peng, J. Wu, and C. K. Liu, "Twist: Teleoperated whole-body imitation system," 2025.

[12] L. Yang, X. Huang, Z. Wu, A. Kanazawa, P. Abbeel, C. Sferrazza, C. K. Liu, R. Duan, and G. Shi, "Omniretarget: Interaction-preserving data generation for humanoid whole-body loco-manipulation and scene interaction," 2025.

[13] Y. Wang, T. Leng, C. Lin, S. Liu, S. Simon, B. Chen, J. Francis, and D. Zhao, "Apex: Learning adaptive high-platform traversal for humanoid robots," 2026.

[14] Z. Wu, X. Huang, L. Yang, Y. Zhang, K. Sreenath, X. Chen, P. Abbeel, R. Duan, A. Kanazawa, C. Sferrazza, G. Shi, and C. K. Liu, "Perceptive humanoid parkour: Chaining dynamic human skills via motion matching," 2026.

[15] D. Wang, X. Wang, C. Zhang, J. Shi, Y. Zhao, C. Bai, and X. Li, "X-loco: Towards generalist humanoid locomotion control via synergetic policy distillation," 2026.

[16] X. Huang, T. Truong, Y. Zhang, F. Yu, J. P. Sleiman, J. Hodgins, K. Sreenath, and F. Farshidian, "Diffuse-cloc: Guided diffusion for physics-based character look-ahead control," *ACM Transactions on Graphics*, vol. 44, no. 4, p. 1–12, Jul. 2025.

[17] G. Tevet, S. Raab, B. Gordon, Y. Shafir, D. Cohen-or, and A. H. Bermano, "Human motion diffusion model," in *The Eleventh International Conference on Learning Representations*, 2023.

[18] G. Tevet, S. Raab, S. Cohan, D. Reda, Z. Luo, X. B. Peng, A. H. Bermano, and M. van de Panne, "Closd: Closing the loop between simulation and diffusion for multi-task character control," 2024.

[19] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

[20] A. Serifi, R. Grandia, E. Knoop, M. Gross, and M. Bächer, "Robot motion diffusion model: Motion generation for robotic characters," in *SIGGRAPH Asia 2024 Conference Papers*, ser. SA '24. New York, NY, USA: Association for Computing Machinery, 2024.

[21] M. Xu, Y. Shi, K. Yin, and X. B. Peng, "Parc: Physics-based augmentation with reinforcement learning for character controllers," in *SIGGRAPH 2025 Conference Papers (SIGGRAPH '25 Conference Papers)*, 2025.

[22] N. Chentanez, M. Müller, M. Macklin, V. Makoviychuk, and S. Jeschke, "Physics-based motion capture imitation with deep reinforcement learning," in *Proceedings of the 11th ACM SIGGRAPH Conference on Motion, Interaction and Games*, ser. MIG '18. New York, NY, USA: Association for Computing Machinery, 2018.

[23] Z. Luo, Y. Yuan, T. Wang, C. Li, S. Chen, F. Castañeda, Z.-A. Cao, J. Li, D. Minor, Q. Ben, X. Da, R. Ding, C. Hogg, L. Song, E. Lim, E. Jeong, T. He, H. Xue, W. Xiao, Z. Wang, S. Yuen, J. Kautz, Y. Chang, U. Iqbal, L. Fan, and Y. Zhu, "Sonic: Supersizing motion tracking for natural humanoid whole-body control," *arXiv preprint arXiv:2511.07820*, 2025.

[24] C. Li, M. Vlastelica, S. Blaes, J. Frey, F. Grimminger, and G. Martius, "Learning agile skills via adversarial imitation of rough partial demonstrations," in *Proceedings of The 6th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, K. Liu, D. Kulic, and J. Ichnowski, Eds., vol. 205. PMLR, 14–18 Dec 2023, pp. 342–352.

[25] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017.

[26] Y. Ze, S. Zhao, W. Wang, A. Kanazawa, R. Duan, P. Abbeel, G. Shi, J. Wu, and C. K. Liu, "Twist2: Scalable, portable, and holistic humanoid data collection system," 2025.

[27] T. He, Z. Luo, W. Xiao, C. Zhang, K. Kitani, C. Liu, and G. Shi, "Learning human-to-humanoid real-time whole-body teleoperation," *arXiv preprint arXiv:2403.04436*, 2024.

[28] D. Hoeller, N. Rudin, D. Sako, and M. Hutter, "Anymal parkour: Learning agile navigation for quadrupedal robots," *Science Robotics*, vol. 9, no. 88, p. eadi7566, 2024.

[29] S. Ross, G. J. Gordon, and J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," 2011.

[30] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. J. Black, "AMASS: Archive of motion capture as surface shapes," in *International Conference on Computer Vision*, Oct. 2019, pp. 5442–5451.

[31] Z. Shen, H. Pi, Y. Xia, Z. Cen, S. Peng, Z. Hu, H. Bao, R. Hu, and X. Zhou, "World-grounded human motion recovery via gravity-view coordinates," in *SIGGRAPH Asia Conference Proceedings*, 2024.

[32] D. D. Team, "Drake: Model-based design and verification for robotics," 2019. [Online]. Available: https://drake.mit.edu

[33] NVIDIA, :, M. Mittal *et al.*, "Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning," 2025.

[34] K. Chen, R. Nemiroff, and B. T. Lopez, "Direct lidar-inertial odometry: Lightweight lio with continuous-time motion correction," *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3983–3989, 2023.

[35] T. Miki, L. Wellhausen, R. Grandia, F. Jenelten, T. Homberger, and M. Hutter, "Elevation mapping for locomotion and navigation using gpu," 2022.

[8] Z. Chen、M. Ji、X. Cheng、X. Peng、X. B. Peng 与 X. Wang，"Gmt：用于人形全身控制的通用运动跟踪"，*arXiv:2506.14770*，2025年。[9] Q. Liao、T. E. Truong、X. Huang、Y. Gao、G. Tevet、K. Sreenath 与 C. K. Liu，"Beyondmimic：从运动跟踪到通过引导扩散实现多功能人形控制"，2025年。[10] M. Ji、X. Peng、F. Liu、J. Li、G. Yang、X. Cheng 与 X. Wang，"Exbody2：先进的表现力人形全身控制"，*arXiv* 预印本 *arXiv:2412.13196*，2024年。[11] Y. Ze、Z. Chen、J. P. Araújo、Z. ang Cao、X. B. Peng、J. Wu 与 C. K. Liu，"Twist：遥操作全身模仿系统"，2025年。[12] L. Yang、X. Huang、Z. Wu、A. Kanazawa、P. Abbeel、C. Sferrazza、C. K. Liu、R. Duan 与 G. Shi，"Omniretarget：面向人形全身移动操作与场景交互的保持交互数据生成"，2025年。[13] Y. Wang、T. Leng、C. Lin、S. Liu、S. Simon、B. Chen、J. Francis 与 D. Zhao，"Apex：学习人形机器人的自适应高平台穿越"，2026年。[14]Z. Wu、X. Huang、L. Yang、Y. Zhang、K. Sreenath、X. Chen、P. Abbeel、R. Duan、A. Kanazawa、C. Sferrazza、G. Shi 与 C. K. Liu，"感知型人形跑酷：通过运动匹配串联动态人类技能"，2026年。[15] D. Wang、X. Wang、C. Zhang、J. Shi、Y. Zhao、C. Bai 与 X. Li，"X-loco：通过协同策略蒸馏迈向通用人形运动控制"，2026年。[16] X. Huang、T. Truong、Y. Zhang、F. Yu、J. P. Sleiman、J. Hodgins、K. Sreenath 与 F. Farshidian，"Diffuse-cloc：用于基于物理的角色前瞻控制的引导扩散"，*ACM* 图形学汇刊，第44卷，第4期，第1–12页，2025年7月。[17] G. Tevet、S. Raab、B. Gordon、Y. Shafir、D. Cohen-or 与 A. H. Bermano，"人体运动扩散模型"，载于 第十一届国际学习表征会议，2023年。[18] G. Tevet、S. Raab、S. Cohan、D. Reda、Z. Luo、X. B. Peng、A. H. Bermano 与 M. van de Panne，"Closd：为多任务角色控制闭合仿真与扩散之间的环路"，2024年。[19] C. Chi、S. Feng、Y. Du、Z. Xu、E. Cousineau、B. Burchfiel 与 S. Song，"Diffusion policy：通过动作扩散进行视觉运动策略学习"，载于 机器人学：科学与系统会议论文集（*RSS*），2023年。[20] A. Serifi、R. Grandia、E. Knoop、M. Gross 与 M. Bächer，"机器人运动扩散模型：面向机器人角色的运动生成"，载于 *SIGGRAPH Asia 2024* 会议论文集，系列 SA '24。美国纽约州纽约市：计算机协会，2024年。[21] M. Xu、Y. Shi、K. Yin 与 X. B. Peng，"Parc：面向角色控制器的基于物理的强化学习增强"，载于 *SIGGRAPH 2025* 会议论文集（*SIGGRAPH '25* 会议论文集），2025年。[22] N. Chentanez、M. Müller、M. Macklin、V. Makoviychuk 与 S. Jeschke，"基于物理的动作捕捉模仿利用深度强化学习，"见 第 *11* 届 *ACM SIGGRAPH* 运动、交互与游戏会议论文集，MIG '18 系列。美国纽约州纽约市：美国计算机协会，2018。[23] Z. Luo、Y. Yuan、T. Wang、C. Li、S. Chen、F. Castañeda、Z.-A. Cao、J. Li、D. Minor、Q. Ben、X. Da、R. Ding、C. Hogg、L. Song、E. Lim、E. Jeong、T. He、H. Xue、W. Xiao、Z. Wang、S. Yuen、J. Kautz、Y. Chang、U. Iqbal、L. Fan 和 Y. Zhu，"Sonic：面向自然人形全身控制的超大规模运动跟踪，"*arXiv* 预印本 *arXiv:2511.07820*，2025。[24] C. Li、M. Vlastelica、S. Blaes、J. Frey、F. Grimminger 和 G. Martius，"通过对抗性模仿粗糙部分演示学习敏捷技能，"见 第六届机器人学习会议论文集，机器学习研究论文集系列，K. Liu、D. Kulic 和 J. Ichnowski 编，第 205 卷。PMLR，2023 年 12 月 14–18 日，第 342–352 页。[25] J. Schulman、F. Wolski、P. Dhariwal、A. Radford 和 O. Klimov，"近端策略优化算法，"2017。[26]Y. Ze、S. Zhao、W. Wang、A. Kanazawa、R. Duan、P. Abbeel、G. Shi、J. Wu 和 C. K. Liu，"Twist2：可扩展、便携且整体的人形数据采集系统，"2025。[27] T. He、Z. Luo、W. Xiao、C. Zhang、K. Kitani、C. Liu 和 G. Shi，"学习人到人形机器人的实时全身遥操作，"*arXiv* 预印本 *arXiv:2403.04436*，2024。[28] D. Hoeller、N. Rudin、D. Sako 和 M. Hutter，"Anymal parkour：为四足机器人学习敏捷导航，"《科学机器人》，第 9 卷，第 88 期，文章 eadi7566，2024。[29] S. Ross、G. J. Gordon 和 J. A. Bagnell，"将模仿学习和结构化预测归约到无悔在线学习，"2011。[30] N. Mahmood、N. Ghorbani、N. F. Troje、G. Pons-Moll 和 M. J. Black，"AMASS：将动作捕捉归档为表面形状，"见 国际计算机视觉会议，2019 年 10 月，第 5442–5451 页。[31] Z. Shen、H. Pi、Y. Xia、Z. Cen、S. Peng、Z. Hu、H. Bao、R. Hu 和 X. Zhou，"通过重力视角坐标实现世界锚定的人体运动恢复，"见 *SIGGRAPH Asia* 会议论文集，2024。[32] D. D. Team，"Drake：面向机器人的基于模型的设计与验证，"2019。 [在线]。可访问：https://drake.mit. edu[33] NVIDIA, :, M. Mittal等，"Isaac lab：一种面向多模态机器人学习的 GPU 加速仿真框架，"2025。[34] K. Chen、R. Nemiroff 和 B. T. Lopez，"直接激光雷达-惯性里程计：具有连续时间运动校正的轻量级 LIO，"*2023 IEEE* 国际机器人与自动化会议（*ICRA*），第 3983–3989 页，2023。[35]T. Miki、L. Wellhausen、R. Grandia、F. Jenelten、T. Homberger 和 M. Hutter，"使用 GPU 进行运动与导航的高程建图，"2022。